

# Assessment Test: Software Tester (QA)

## Task Management Application — Complete Test Case & Bug Assessment

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## 1. Test Case Design

### A. Registration

| TC ID  | Test Scenario                      | Test Data / Steps                                      | Expected Result                                  | Type     |
|--------|------------------------------------|--------------------------------------------------------|--------------------------------------------------|----------|
| REG-01 | Register with valid details        | Enter valid name, unique email and valid password      | User should be registered successfully           | Positive |
| REG-02 | Register with existing email       | Enter an already registered email                      | Proper error should be displayed                 | Negative |
| REG-03 | Empty registration fields          | Leave all fields blank and submit                      | Mandatory field validation should appear         | Negative |
| REG-04 | Invalid email format               | Enter abc@ or abc.com                                  | Invalid email message should appear              | Negative |
| REG-05 | Password below minimum length      | Enter a very short password                            | Password validation should be displayed          | Edge     |
| REG-06 | Password at minimum allowed length | Enter password exactly at minimum limit                | Registration should work if it meets requirement | Edge     |
| REG-07 | Special characters in name         | Enter valid name containing allowed special characters | Application should handle input correctly        | Edge     |
| REG-08 | Uppercase email                    | Enter USER@EMAIL.COM                                   | System should correctly handle email             | Edge     |

### B. Login

| TC ID    | Test Scenario                | Test Data / Steps                   | Expected Result                                 | Type     |
|----------|------------------------------|-------------------------------------|-------------------------------------------------|----------|
| LOGIN-01 | Login with valid credentials | Registered email + correct password | User should login successfully                  | Positive |
| LOGIN-02 | Wrong password               | Valid email + incorrect password    | Error message should be displayed               | Negative |
| LOGIN-03 | Unregistered email           | Email not registered                | Login should fail with proper message           | Negative |
| LOGIN-04 | Both fields blank            | Submit without entering data        | Required field validation should appear         | Negative |
| LOGIN-05 | Invalid email format         | Enter invalid email                 | Email validation should appear                  | Negative |
| LOGIN-06 | Password case sensitivity    | Enter password with incorrect case  | Login should fail if password is case-sensitive | Edge     |
| LOGIN-07 | Multiple failed attempts     | Enter wrong password repeatedly     | Application should handle attempts securely     | Edge     |

### C. Task CRUD Operations

| TC ID   | Test Scenario                     | Test Data / Steps           | Expected Result                         | Type     |
|---------|-----------------------------------|-----------------------------|-----------------------------------------|----------|
| TASK-01 | Create valid task                 | Enter valid title/details   | Task created and stored in database     | Positive |
| TASK-02 | Create with mandatory field blank | Leave required field blank  | Validation message should appear        | Negative |
| TASK-03 | Maximum allowed characters        | Enter data at maximum limit | Task should be created if within limit  | Edge     |
| TASK-04 | Exceed character limit            | Enter data beyond limit     | Proper validation should appear         | Edge     |
| TASK-05 | Special characters                | Enter special characters    | Input should be handled correctly       | Edge     |
| TASK-06 | View task list                    | Open task list after login  | User's tasks should be displayed        | Positive |
| TASK-07 | New task appears                  | Create task then view list  | New task should appear correctly        | Positive |
| TASK-08 | No tasks                          | User has no tasks           | Empty-state message should be displayed | Edge     |

|         |                                |                                       |                                           |          |
|---------|--------------------------------|---------------------------------------|-------------------------------------------|----------|
| TASK-09 | Refresh task list              | Refresh page                          | Saved tasks should remain available       | Positive |
| TASK-10 | Task ownership                 | Attempt to access another user's task | Unauthorized access should not be allowed | Negative |
| TASK-11 | Edit valid task                | Edit existing task with valid data    | Task should update successfully           | Positive |
| TASK-12 | Edit with blank required field | Clear required field and save         | Validation should appear                  | Negative |
| TASK-13 | Edit at maximum data           | Use maximum allowed data              | Task should save successfully             | Edge     |
| TC ID   | Test Scenario                  | Test Data / Steps                     | Expected Result                           | Type     |
| TASK-14 | Cancel edit                    | Change data and cancel                | Original task data should remain          | Positive |
| TASK-15 | Delete existing task           | Delete a task                         | Task removed from list/database           | Positive |
| TASK-16 | Cancel deletion                | Cancel confirmation                   | Task should not be deleted                | Positive |
| TASK-17 | Delete then refresh            | Delete task and refresh               | Deleted task should not reappear          | Positive |
| TASK-18 | Delete another user's task     | Attempt unauthorized deletion         | Unauthorized deletion should be blocked   | Negative |

## D. Input Validation & Error Handling

| TC ID  | Test Scenario                  | Expected Result                                        |
|--------|--------------------------------|--------------------------------------------------------|
| VAL-01 | Blank mandatory fields         | Clear validation messages should appear                |
| VAL-02 | Invalid email                  | System should reject invalid email                     |
| VAL-03 | Extremely long input           | Application should safely handle/reject input          |
| VAL-04 | Special/script-like input      | Input should be safely handled and not execute as code |
| VAL-05 | Database/server error          | User should receive a proper error message             |
| VAL-06 | Network failure while creating | Application should not show false success              |
| VAL-07 | Invalid login credentials      | Clear login error should be displayed                  |
| VAL-08 | Session expiry                 | User should be redirected/asked to login               |
| VAL-09 | Duplicate registration         | User should be informed account already exists         |
| VAL-10 | Refresh after CRUD             | Data should remain consistent with database            |

## 2. Potential Bugs / Risk Areas

### Potential Bugs / Risk Areas

| #  | Bug Description                                             | Severity | Reason / Impact                                                 |
|----|-------------------------------------------------------------|----------|-----------------------------------------------------------------|
| 1  | User can view another user's tasks                          | Critical | Major privacy/security issue; confidential data may be exposed. |
| 2  | User can delete another user's task by manipulating task ID | Critical | Unauthorized data deletion.                                     |
| 3  | Password stored/transmitted insecurely                      | Critical | Could expose credentials and compromise accounts.               |
| 4  | Duplicate email registration is allowed                     | Major    | Can create account/data inconsistency.                          |
| 5  | Created task is not saved correctly in database             | Major    | User may lose task data.                                        |
| 6  | Edited task shows old data after refresh                    | Major    | UI and database data become inconsistent.                       |
| 7  | Deleted task appears again after refresh                    | Major    | Indicates delete/database synchronization problem.              |
| 8  | Application crashes when mandatory fields are empty         | Major    | Poor error handling can block users.                            |
| 9  | Very long/special input breaks task form                    | Major    | Can cause UI/security issues if not safely handled.             |
| 10 | Session does not expire/logout correctly                    | Major    | Unauthorized person on same device may access account/tasks.    |

## 3. Testing Approach

1. Registration & Login — verify secure account creation and authentication.
2. Authorization/Security — verify users can access only their own tasks.
3. CRUD — verify Create, Read/View, Update/Edit and Delete operations.
4. Database Persistence — verify data remains correct after refresh and re-login.
5. Input Validation — verify invalid and boundary inputs are handled correctly.
6. Error Handling — verify failures produce clear messages without crashes.
7. Edge Cases — verify maximum lengths, empty data, special characters and repeated actions.

## 4. Final QA Acceptance Criteria

- Valid users can register and login successfully.
- Invalid credentials are rejected.
- Users can perform all task CRUD operations correctly.
- Task data is correctly persisted in the database.
- Users cannot access or modify other users' tasks.
- Mandatory fields and invalid inputs are properly validated.
- Errors are displayed clearly without application crashes.
- Deleted/updated data remains consistent after refresh and re-login.